

## Multiple Choice (10 marks)

1. In Python 3, let  $r = \text{lambda } q: q * 2$ . What is  $r(3)$ ?
  - (a) 2
  - (b) 6
  - (c) 3
  - (d) 1
2. Which of the following statements about a remote procedure call is true?
  - (a) It is used to call procedures with addresses that are farther than  $2^{16}$  bytes away.
  - (b) It cannot return a value.
  - (c) It cannot pass parameters by reference.
  - (d) It cannot call procedures implemented in a different language.
3. In Python 3, which of the following function checks in a string that all characters are in uppercase?
  - (a) `isupper()`
  - (b) `join(seq)`
  - (c) `len(string)`
  - (d) `ljust(width[, fillchar])`
4. What is the value of this python expression:  $4 + 3 \% 2$ ?
  - (a) 1
  - (b) 6
  - (c) 4
  - (d) 5
5. In python 3, which of the following is floor division?
  - (a) `/`
  - (b) `//`
  - (c) `%`
  - (d) `|`

## True / False (10 marks)

*Answer True or False*

6. True or False: Python variable names are case-sensitive, meaning 'Variable' and 'variable' are treated as distinct identifiers.
7. True or False: In a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes is approximately  $1/K$ .
8. True or False: The minimum value in the list  $l = [1,2,3,4]$  in Python 3 is 2.
9. True or False: The string 'xxzy' can be generated by the provided grammar, as it matches the recursive structure of the rules.
10. True or False: The value of the Python expression  $1 + 3 \% 3$  is 1.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to divide two lists using map and lambda function.
12. Write a function to sort a list of tuples in increasing order by the last element in each tuple.

*End of Paper*